The Perfect Feedback Exponent is Not Achieved Everywhere by Every Asymptoting Two Phase Noisy Feedback Scheme

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Abstract

In this brief note, we present a result pertaining to the bounds on the error exponent of a continuous time gaussian channel with noisy feedback.

In the SM thesis [1], a simple two phase scheme is presented for the noisy gaussian feedback setting in continuous time. The achieved reliability function is lower bounded there. Here we make use of a bound from [2, 3]. It is an upper bound on the perfect feedback reliability function.

Figure 1 shows both the bounds plotted on the same graph. for different values of the feedback capacity (increasing from top to bottom, with perfect feedback in the bottom-most subplot) and a fixed, large value of the Cprime parameter of the forward channel. Figure 2 shows the same situation but with a small value of the Cprime parameter of the forward channel.

The figures suggest that, depending upon the value of Cprime, the rudimentary two phase scheme of [1] performs optimally only for low rates. What makes this come about is that the pre-factor multiplying $(1 - \frac{r}{C_{forward}})$ is $(\frac{1}{4C_{forward}} + \frac{1}{C_{feedback}})^{-1}$. The first term in the inverse sum ensures that $C_{feedback}$ cannot have an unconstrained effect.

References

[1] Aman Chawla. Reliability of a gaussian channel in the presence of gaussian feedback. *S.M. Thesis*, 2006.

[2] Peter Berlin, Barış Nakiboğlu, Bixio Rimoldi, and Emre Telatar. A simple converse of burnashev's reliability function. *Information Theory, IEEE Transactions on*, 55(7):3074–3080, 2009.

[3] Aman Chawla and Salvatore Domenic Morgera. On the continuous time additive gaussian noise channel in the presence of perfect feedback. In *2020 IEEE Information Theory Workshop (ITW)*, pages 1–5. IEEE, 2021.

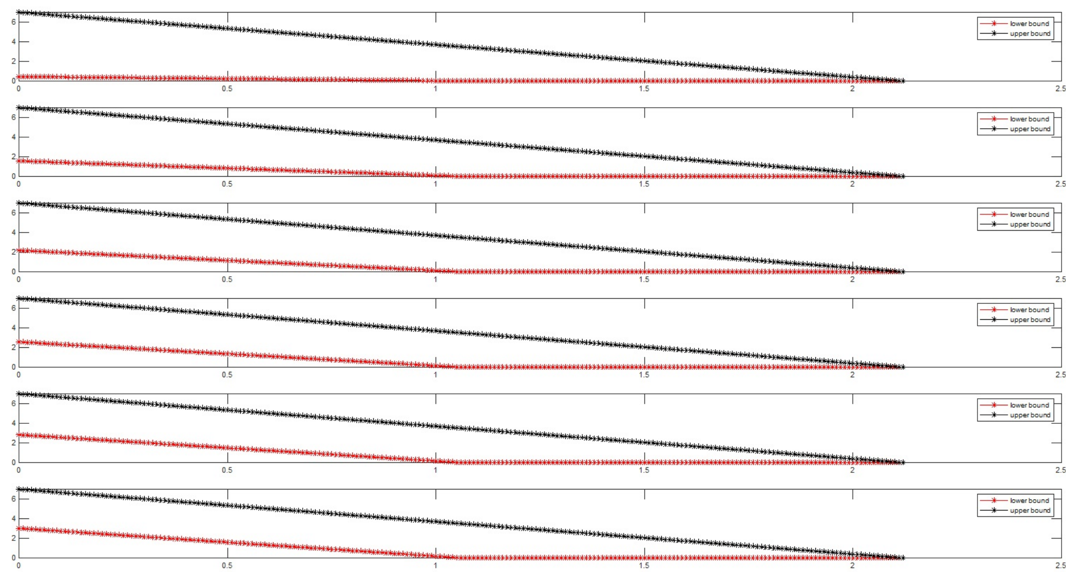

Figure 1: High value of Cprime. The horizontal axis of each sub-plot is the average transmission rate, and the vertical axes represent the error exponent bounds. The forward channel capacity is taken as 1.06 bits per second.

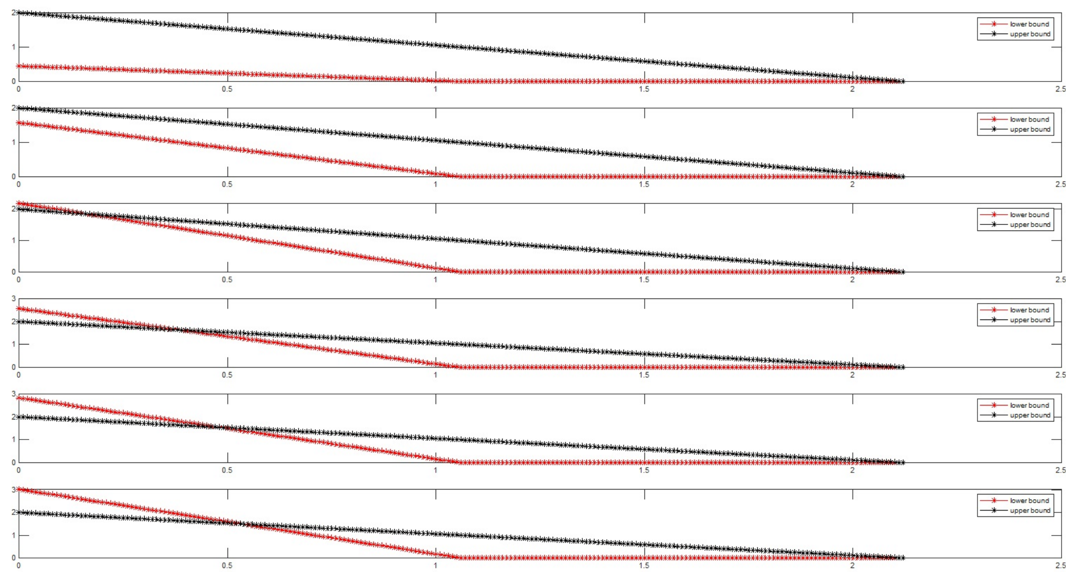

Figure 2: Low value of Cprime. The horizontal axes are the average transmission rate and the vertical axes represent the error exponent bounds. Where the lower bound (red) crosses above the upper bound (black), the upper bound indicates the exponent achieved. The forward channel capacity is taken as 1.06 bits per second.